

014 – E2E Test Coverage Discovery (R9)

Purpose: Complete coverage map of all UI operations discovered by live-browser exploration.

Maps every user journey to a test classification, data strategy, and key locators before any fixture-managed or isolated-sequence tests are written.

Status: In Progress — initial exploration complete; under review; additional MCP scenarios may be added.

Last Updated: 2026-05-17

Exploration method: MCP Playwright browser tools against the live development stack

(<http://localhost:5175>).

Exploration state: Mode 2 — site **POT E2E Site**, two canonical E2E users, one account "Main Checking", one expense "Monthly Rent", one income "Monthly Salary".

1. Route Inventory

All routes confirmed from [src/routes/AppRoutes.tsx](#) and live navigation.

Route	Type	Notes
/login	Public	Login, Forgot Password, Sign Up dialogs
/logout	Public	Triggers logout + redirects to /login
/	Protected	Redirects to /dashboard
/dashboard	Protected	Quick Actions + overview sections
/projections	Protected	Chart view with view selector, period, date, account filter
/accounts	Protected	Accounts table
/accounts/create	Protected overlay	Sheet rendered as child route
/accounts/edit/:id	Protected overlay	Sheet rendered as child route
/expenses	Protected	Expenses table with search + account filter
/expenses/create	Protected overlay	Sheet rendered as child route
/expenses/create?duplicate=:id	Protected overlay	Pre-filled duplicate sheet
/expenses/edit/:id	Protected overlay	Sheet rendered as child route
/incomes	Protected	Income table with search + account filter

Route	Type	Notes
/incomes/create	Protected overlay	Sheet rendered as child route
/incomes/edit/:id	Protected overlay	Sheet rendered as child route
/users	Protected	Users table
/users/invite	Protected overlay	Sheet rendered as child route
/approvals/pending	Protected	Platform admin only; not in sidebar for non-platform-admin
* (unknown)	Protected catch-all	Redirects to /dashboard

Route guard behaviour:

- Unauthenticated access to any protected route → redirect to [/login](#)
- Unknown route when authenticated → redirect to [/dashboard](#)
- Root / → redirect to [/dashboard](#)

2. Sidebar Navigation

Sidebar items confirmed from [src/components/nav/AppSidebarMenus.tsx](#).

Label	Href/Action	Required Permission	Admin Visible	Viewer Visible
Dashboard	/dashboard	(authenticated)	✓	✓
Projections	/projections	(authenticated)	✓	✓
Accounts	/accounts	(authenticated)	✓	✓
Expenses	/expenses	(authenticated)	✓	✓
Income	/incomes	(authenticated)	✓	✓
Users	/users	user:manage or user:view	✓	✓
Approvals	/approvals/pending	platform:manage	✓ (platform admin only)	X
Export...	Opens dialog	maintenance:export	✓	X
Import...	Opens dialog	maintenance:import	✓	X

Mobile behaviour: Export and Import are hidden at mobile viewport (`isMobile === true`).

Sidebar toggle: At mobile viewport ($\leq \sim 768\text{px}$) the sidebar is collapsed by default. [Toggle Sidebar](#) button

is visible in the header; clicking it opens the sidebar overlay.

3. Authentication / Login Page

3.1 Login Form

Operation	Expected Outcome	Classification
Load <code>/login</code>	Username + Password fields visible, Login button, "Forgot your password?" link, "Sign up" link	parallel-safe
Submit empty form	Login button disabled — cannot submit	parallel-safe
Submit with wrong credentials	Inline error "Invalid username or password" in form (generic div, no <code>[role="alert"]</code> , no toast)	parallel-safe (transient session)
Submit with valid credentials	Redirect to <code>/dashboard</code> , refresh token cookie <code>pot_refresh_token</code> set	parallel-safe (transient session)

Login form locators:

- Username: `page.getByLabel('Username')`
- Password: `page.getByLabel('Password')`
- Submit: `page.getByRole('button', { name: 'Login' })`
- Inline error: element within the login form containing the error text (no stable role/testid yet — record as locator gap)

3.2 Forgot Password Dialog

Operation	Expected Outcome	Classification
Click "Forgot your password?"	Dialog opens with title "Reset Password", username field, "Send Reset Code" button	parallel-safe
"Send Reset Code" with empty field	Button disabled — cannot submit	parallel-safe
"Send Reset Code" with username filled	Step 2 (not yet explored; expected: code + new password fields)	—

Forgot Password locators:

- Trigger: `page.getByRole('button', { name: 'Forgot your password?' })` (rendered as `<button>`)
- Dialog: `page.getByRole('dialog')`
- Username: `page.getByLabel('Username')` scoped inside dialog
- Send button: `page.getByRole('button', { name: 'Send Reset Code' })`

3.3 Sign Up Dialog

Operation	Expected Outcome	Classification
Click "Sign up"	Dialog opens with title "Create Account", Username + Email fields, "Send Verification Code" button	parallel-safe
"Send Verification Code" with empty fields	Button disabled	parallel-safe
Fill username + email → Send Verification Code	Step 2 (not yet explored; expected: OTP/code entry field)	—

Sign Up locators:

- Trigger: `page.getByRole('button', { name: 'Sign up' })`
- Dialog: `page.getByRole('dialog')`
- Username: `page.locator('[name="username"]')` scoped inside dialog
- Email: `page.locator('[name="email"]')` scoped inside dialog
- Send button: `page.getByRole('button', { name: 'Send Verification Code' })`

Note: Multi-step Sign Up (step 2: verification code entry) and multi-step Password Reset (step 2: code + new password) were not fully explored. These flows require a real email or a test OTP strategy. Defer until OTP testing strategy is decided (see PRD D7).

4. Dashboard

4.1 Quick Actions (Admin only)

Operation	Expected Outcome	Classification
Load dashboard as admin	"Quick Actions" section visible with 4 cards: Income Renewal, Expense Accrual, Budget Reminder, Income Confirmation	parallel-safe
Load dashboard as viewer	"Quick Actions" section not present	parallel-safe
Click "Income Renewal" card	Toast <code>[data-sonner-toast]</code> with text containing "Income Renewal"	isolated-sequence
Click "Expense Accrual" card	Toast with text containing "Expense Accrual"	isolated-sequence
Click "Budget Reminder" card	Toast (email trigger)	isolated-sequence
Click "Income Confirmation" card	Toast with text containing "Income Confirmation"	isolated-sequence

Quick Action locators:

- Quick Actions heading: `page.getByRole('heading', { name: 'Quick Actions' })`
- Individual cards: generic divs. Locate via heading text inside card. Cannot use `getByRole('button')`.

- Click pattern: `page.evaluate(() => document.querySelector('[text*="Income Renewal"]')?.click())` or scope by heading within section
- Toast: `page.locator('[data-sonner-toast]')`

Testability gap: Quick Action cards have no accessible role or `data-testid`. Recommend adding `role="button"` or `data-testid="quick-action-{id}"` to each card for reliable E2E targeting.

4.2 Overview Sections

Section	Viewer	Admin	Classification
"Accounts Overview" collapsible section	✓	✓	parallel-safe (with seeded data)
"Expenses Overview" collapsible section (30-day)	✓	✓	parallel-safe (with seeded data)
"Income Overview" collapsible section (30-day)	✓	✓	parallel-safe (with seeded data)
"\$0.00" empty-state values	✓	✓	parallel-safe (baseline DB with no financial data)

Collapse/expand locators:

- Section toggle buttons: `page.getByRole('button', { name: 'Accounts Overview' })` etc.

5. Accounts Page

5.1 Page State

Operation	Expected Outcome	Classification
Load <code>/accounts</code> (empty)	Empty state message visible, "Add Account" button present (admin)	parallel-safe
Load <code>/accounts</code> (with data)	Table with columns: Account Name, BSB, Account Number, Balance, Reserved, Actions	parallel-safe (with seeded data)
Column headers visible	Account Name, BSB, Account Number, Balance, Reserved	parallel-safe

Toolbar locators:

- Add button: `page.getByRole('button', { name: /add.*account/i })` or `page.getByLabel('Add a new account')`

5.2 Create Account

Operation	Expected Outcome	Classification
Click "Add Account"	Navigates to <code>/accounts/create</code> , sheet/dialog opens with title "Create Account"	fixture-managed

Operation	Expected Outcome	Classification
Submit empty form	Validation errors on required fields	fixture-managed
Fill valid data + Submit	Account created, redirect to <code>/accounts</code> , row appears in table, toast success	fixture-managed

Create form fields (confirmed via live exploration):

- Account Name (text, required)
- BSB (text, format `000-000`)
- Account Number (text)
- Opening Balance (number)
- Reserved Amount (number)

Form locators:

- Dialog: `page.getByRole('dialog')`
- Form fields: `page.getByLabel('Account Name')` etc. scoped inside dialog

5.3 Edit Account

Operation	Expected Outcome	Classification
Click row "Open menu" → "Edit"	Navigates to <code>/accounts/edit/:id</code> , sheet/dialog opens pre-filled	fixture-managed
Modify a field + Submit	Account updated, redirect to <code>/accounts</code> , row updated in table, toast success	fixture-managed
Viewer clicks "Edit"	Edit button disabled (<code>aria-disabled="true"</code>)	parallel-safe

5.4 Delete Account

Operation	Expected Outcome	Classification
Click row "Open menu" → "Delete"	Confirmation dialog appears ("Are you sure?")	fixture-managed
Confirm deletion	Account removed from table, toast success	fixture-managed
Viewer sees "Delete"	Menu item disabled (<code>aria-disabled="true"</code>)	parallel-safe

Row action locators:

- Open menu: `page.getByRole('button', { name: 'Open menu' })` scoped to row
- Menu item: `page.getByRole('menuitem', { name: 'Edit' })` / `page.getByRole('menuitem', { name: 'Delete' })`
- Confirmation dialog: `page.getByRole('dialog')`
- Confirm button: `page.getByRole('button', { name: 'Delete' })` inside dialog

6. Expenses Page

6.1 Page State

Operation	Expected Outcome	Classification
Load <code>/expenses</code> (empty, no accounts)	Empty state, "Add Expense" button disabled with tooltip "Create at least one account first"	parallel-safe
Load <code>/expenses</code> (empty, accounts exist)	Empty state, "Add Expense" button enabled	parallel-safe (with seeded data)
Load <code>/expenses</code> (with data)	Table with columns: Description, Account, Category(?), Amount, Due Date, Frequency, Paid Date, Status, Actions	parallel-safe (with seeded data)
Search by description	Filters table rows, <code>FilterResultsCount</code> updates	fixture-managed
Account filter (multi-account)	Filter dropdown appears only when <code>accountsInItems.length > 1</code> , sets <code>?accountId=</code> URL param	fixture-managed

Search locator: `page.getByLabel('Search expenses by description')` (from `ariaLabel="Search expenses by description"`)

Account filter locator: `page.getByLabel('Filter by account')` (Radix Select, NOT native `<select>`)

Account filter note: Filter is only rendered when items span more than one account. Testing requires at least two accounts with expenses. URL state: `?accountId={id}`.

6.2 Create Expense

Operation	Expected Outcome	Classification
Click "Add Expense"	Navigates to <code>/expenses/create</code> , sheet slides in from right	fixture-managed
Submit empty form	Validation errors on required fields	fixture-managed
Fill valid data + Submit	Expense created, sheet closes, row appears in table, toast success	fixture-managed

Create form fields (confirmed via live exploration):

- Description (text, required)
- Account (Radix combobox, required) — click trigger → `[role="option"]` items
- Amount (number, required)
- Due Date (custom date picker — click button → calendar popover → Today / Accept / Cancel)
- Frequency Number (number) + Frequency Period (Radix combobox: days/weeks/months/years)
- Category (Radix combobox or text?)
- Accrual fields (Accrual Mode Radix combobox — "None", "Advance", "Arrears"; conditionally reveals additional fields)
- Notes / Tags (if present — not fully confirmed)

Accrual field note: When Accrual Mode $\neq$ "None", additional fields appear (accrual amount, accrual period). Requires two sub-tests (no-accrual and with-accrual paths).

6.3 Row Actions (Expense)

Action	Expected Outcome	Classification
"Mark as Paid"	Paid Date column populates, row status changes, toast success	fixture-managed
"Toggle Exclusion" (exclude)	Row gains visual indicator (<code>border-l-4 border-l-slate-400</code>), "Excluded" badge appears in row, toast	fixture-managed
"Toggle Exclusion" (re-include)	Visual indicator removed, toast	fixture-managed
"Edit"	Navigates to <code>/expenses/edit/:id</code> , sheet opens pre-filled	fixture-managed
"Duplicate"	Navigates to <code>/expenses/create?duplicate=:id</code> , sheet opens pre-filled with same data	fixture-managed
"Delete"	Confirmation dialog; on confirm, row removed + toast	fixture-managed
Viewer row actions	"Open menu" button not present / all actions disabled	parallel-safe

Row action locators:

- Open menu: `page.getByRole('button', { name: 'Open menu' })` scoped to row
- Actions: `page.getByRole('menuitem', { name: 'Mark as Paid' })` etc.
- Excluded row: `page.locator('tr').filter({ has: page.getByText('Excluded') })`

7. Income Page

7.1 Page State

Operation	Expected Outcome	Classification
Load <code>/incomes</code> (empty, no accounts)	Empty state, "Add Income" button disabled	parallel-safe
Load <code>/incomes</code> (with data)	Table with columns: Description, Account, Amount, Expected Date, Frequency, Received Date, Status, Actions	parallel-safe (with seeded data)
Search by description	Filters table rows	fixture-managed
Account filter	Same behaviour as Expenses (multi-account only, <code>?accountId=</code> URL param)	fixture-managed

Search locator: `page.getByLabel('Search incomes by description')` (expected — mirror of expenses pattern)

Add button: `page.getByLabel('Add a new income')`

7.2 Create Income

Operation	Expected Outcome	Classification
Click "Add Income"	Navigates to <code>/incomes/create</code> , sheet opens	fixture-managed
Submit empty form	Validation errors	fixture-managed
Fill valid data + Submit	Income created, row appears in table, toast success	fixture-managed

Create form fields (confirmed via live exploration — no accrual fields):

- Description (text, required)
- Account (Radix combobox, required)
- Amount (number, required)
- Expected Date (custom date picker)
- Frequency Number + Frequency Period (Radix combobox)

7.3 Row Actions (Income)

Action	Expected Outcome	Classification
"Mark as Received" (unpaid)	Received Date populates, row status changes, toast	fixture-managed
"Mark as Paid" (received state variant — label may differ)	Reverts received status or confirms payment	fixture-managed
"Edit"	Navigates to <code>/incomes/edit/:id</code> , sheet opens pre-filled	fixture-managed
"Duplicate"	Navigates to <code>/incomes/create?duplicate=:id</code> , sheet opens pre-filled	fixture-managed
"Delete"	Confirmation dialog; on confirm row removed + toast	fixture-managed
Viewer row actions	All disabled	parallel-safe

Note: Income row action label for marking received may be "Mark as Received" (not "Mark as Paid"). Confirm exact label from live data.

8. Users Page

8.1 Page State

Operation	Expected Outcome	Classification
Load <code>/users</code> as admin	Table with columns: Username, Email, Status, Last Login, Roles, Actions	parallel-safe
Load <code>/users</code> as viewer	Table visible, row action menus absent	parallel-safe

8.2 Change Role

Operation	Expected Outcome	Classification
Admin clicks "Open menu" → "Change Role"	Dialog opens with title "Change Role", role options listed	fixture-managed
Select role + Submit	User role updated, row updated, toast	fixture-managed

8.3 Disable / Enable User

Operation	Expected Outcome	Classification
Admin clicks "Disable User"	Immediate action (no confirmation dialog), user status → disabled, toast	fixture-managed
Admin clicks "Enable User"	Immediate action, user status → enabled, toast	fixture-managed

8.4 Invite User

Operation	Expected Outcome	Classification
Admin clicks "Invite User"	Navigates to <code>/users/invite</code> , sheet opens with invite form	fixture-managed
Submit empty form	Validation errors on required fields	fixture-managed
Fill valid data + Submit	Invite sent/user created, toast; sheet closes	fixture-managed
Viewer "Invite User" button	Button disabled (<code>aria-disabled</code>)	parallel-safe

Invite form fields (confirmed via live exploration):

- Username (text, required)
- Email (text, required)
- Role (Radix combobox — "Admin" / "Viewer")

Validation observed:

- Username already taken → form-level error
- Invalid email → form-level error

9. Projections Page

9.1 View and Filter Controls

Operation	Expected Outcome	Classification
Load <code>/projections</code>	Chart visible, view selector (5 options), period buttons, date picker, account filter	parallel-safe (empty data) / isolated-sequence (seeded data)
Change view selector option	Chart re-renders for selected view	isolated-sequence
Click period button (e.g., "30 days")	Chart updates for selected period	isolated-sequence
Change date via date picker	Chart shifts to selected start date	isolated-sequence
Select account from account filter	Chart filters to that account	isolated-sequence

View selector options (Radix combobox, confirmed via live exploration):

- Option 1–5 (exact labels not captured — recommend re-checking source)

Locators:

- Account filter: `page.getByLabel('Filter by account')`
- Period buttons: `page.getByRole('button', { name: /30 days|60 days|90 days|6 months|1 year/i })`
- Date picker: `page.getByRole('button', { name: /\d{4}/ })` (button displaying current date)

10. Export Dialog

Operation	Expected Outcome	Classification
Click "Export..." in sidebar	Dialog opens with single export confirmation button	isolated-sequence
Confirm export	<code>.export</code> file downloaded, dialog closes, toast	isolated-sequence
Viewer has no "Export..."	Sidebar item absent	parallel-safe

Export locators:

- Sidebar link: `page.getByRole('link', { name: 'Export...' })` (or button via `onClick` sidebar item)
- Dialog: `page.getByRole('dialog')`
- Confirm button: `page.getByRole('button', { name: /export/i })` inside dialog

11. Import Dialog

Operation	Expected Outcome	Classification
Click "Import..." in sidebar	Dialog opens with "Browse" file picker + "Import Data" button	isolated-sequence
Select valid <code>.export</code> file + Import Data	Import completes, toast success, data visible	isolated-sequence
Select invalid file + Import Data	Error feedback (toast or inline error)	isolated-sequence
Viewer has no "Import..."	Sidebar item absent	parallel-safe

Import locators:

- Sidebar link: `page.getByRole('link', { name: 'Import...' })` (or `onClick` item)
- Dialog: `page.getByRole('dialog')`
- File input: `page.locator('input[type="file"]')` inside dialog
- Import button: `page.getByRole('button', { name: 'Import Data' })`

12. Settings Dialog (User Dropdown)

Operation	Expected Outcome	Classification
Click user display name (top of sidebar)	Dropdown menu with "Settings", "Log Out"	parallel-safe
Click "Settings"	Dialog opens with accordion sections	parallel-safe

Settings dialog sections (confirmed via live exploration):

1. **Site Details** — Display Name, Site Name fields (admin only? confirm permission)
2. **User Details** — username/email display or edit fields
3. **Change Password** — Current Password, New Password, Confirm Password
4. **Budget Reminders** — toggle/config for reminder emails

Section	Operation	Classification
Site Details	Update site display name	fixture-managed
User Details	Update user profile (username/email?)	fixture-managed
Change Password	Submit valid new password	fixture-managed
Change Password	Submit mismatched new/confirm	parallel-safe (no state change on failure)
Budget Reminders	Toggle on/off	fixture-managed

Settings locators:

- User menu trigger: `page.getByRole('button', { name: 'e2e_admin' })` (or user display name)

- Settings item: `page.getByRole('menuitem', { name: 'Settings' })`
- Dialog: `page.getByRole('dialog')`
- Accordion: `page.getByRole('button', { name: 'Site Details' })` etc.

13. Pending Approvals Page

Operation	Expected Outcome	Classification
Admin navigates to <code>/approvals/pending</code>	Page renders, <code>GET /api/approvals/pending</code> succeeds (empty list when no pending approvals)	parallel-safe
Viewer: "Approvals" not in sidebar	Sidebar item absent for viewer role	parallel-safe
Unauthenticated access	Redirect to <code>/login</code>	parallel-safe

Note (potentially resolved): During discovery, `GET /api/approvals/pending` returned 403 for `e2e_admin` because the user's RowId (`58d1a6b2-bf25-40c1-b3d6-656ccf4a68b5`) was absent from the `PlatformAdmin:UserIds` configuration. Fix applied: RowId added to `PLATFORM_ADMIN_USERIDS` in `.env.development` and `--PlatformAdmin:UserIds` in `playwright.config.ts`. The Approvals sidebar item should now appear for `e2e_admin` and the API call should succeed.

14. Role Comparison Summary

Feature Area	Admin	Viewer
Quick Actions (dashboard)	✓ visible + clickable	X not rendered
Add / Edit / Delete Accounts	✓ buttons enabled	X disabled + <code>aria-disabled="true"</code>
Add / Edit / Delete Expenses	✓ enabled	X disabled
Add / Edit / Delete Income	✓ enabled	X disabled
Row action menus (Expenses/Income)	✓	X menus not rendered
Invite User	✓ enabled	X disabled
User row action menus	✓	X not rendered
Export... sidebar item	✓	X absent
Import... sidebar item	✓	X absent
Approvals sidebar item	✓ (platform admin only)	X absent
Settings dialog	✓	✓ (some sections may differ)
Projections (read)	✓	✓

Feature Area	Admin	Viewer
Accounts / Expenses / Income (read)	✓	✓
Users table (read)	✓	✓

Viewer disabled-control pattern:

- `<button>`: `disabled` attribute + `aria-disabled="true"` → `.isDisabled()` returns true
- Menu items: `data-disabled=""` + `aria-disabled="true"` → cannot be clicked without `{ force: true }`

15. Mobile / Responsive Behaviour

Behaviour	Detail
Sidebar at $\leq$ ~768px wide viewport	Collapsed by default; nav links not visible
Toggle Sidebar button	Visible in header when sidebar is collapsed; click reveals sidebar overlay
Export / Import sidebar items	Hidden at mobile viewport regardless of permissions
After clicking Toggle Sidebar	Sidebar opens and links become visible again

Mobile locators:

- Toggle button: `page.getByRole('button', { name: 'Toggle Sidebar' })`
- After toggle: `page.getByRole('link', { name: 'Accounts' })` etc.

16. Key Locator Reference

Element	Recommended Locator
Toast notification	<code>page.locator('[data-sonner-toast]')</code>
Form field validation error	<code>page.locator('[data-slot="form-message"]')</code>
Modal dialog	<code>page.getByRole('dialog')</code>
Sheet (slide-in panel)	<code>page.locator('[data-slot="sheet-content"]')</code>
Row open-menu button	<code>page.getByRole('button', { name: 'Open menu' })</code> (scoped to row)
Menu item (inside open menu)	<code>page.getByRole('menuitem', { name: '...' })</code>
Sidebar nav link	<code>page.getByRole('link', { name: '...' })</code>
Custom combobox (Radix)	Click trigger text → <code>page.getByRole('option', { name: '...' })</code>

Element	Recommended Locator
Custom date picker	Click button with date text → calendar appears with Today / Cancel / Accept
Account filter combobox	<code>page.getByLabel('Filter by account')</code>
Search input (expenses)	<code>page.getByLabel('Search expenses by description')</code>
Search input (income)	<code>page.getByLabel('Search incomes by description')</code>
Clear search button	<code>page.getByLabel('Clear search input')</code>
Add account button	<code>page.getByLabel('Add a new account')</code>
Add expense button	<code>page.getByLabel('Add a new expense')</code>
Add income button	<code>page.getByLabel('Add a new income')</code>
Toggle sidebar button	<code>page.getByRole('button', { name: 'Toggle Sidebar' })</code>
POT logo / home link	<code>page.getByLabel('Navigate to Pay On Time homepage')</code>
Error boundary alert	<code>page.getByRole('alert')</code> (full-screen fallback)

17. Validation Patterns

Pattern	Details
Login — invalid credentials	<code>[role="alert"]</code> element inside form containing the error text
Create/Edit forms — required fields	<code>[data-slot="form-message"]</code> elements appear below each invalid field
Create/Edit forms — submit with errors	Form stays open; error messages visible
Invite User — duplicate username	Form-level error (confirmed observed)
Password change — mismatch	Form-level error on "Confirm Password" field
Toast — success operations	<code>[data-sonner-toast]</code> with descriptive text (e.g., "Account created", "Income Renewal Complete")
Disabled controls	<code>aria-disabled="true"</code> on both buttons and Radix menu items

Resolved: `role="alert"` added to the login inline error element in `LoginForm.tsx`. Use `page.getByRole('alert')` scoped within the login card for reliable assertion.

18. Test Classification Summary

18.1 Parallel-Safe Tests (no state change)

- Login page renders (fields, buttons, links visible)
- Login submit disabled when fields empty
- Login with wrong credentials → inline error (transient session, context closed immediately)
- Forgot Password dialog opens, Send Reset Code disabled when empty
- Sign Up dialog opens, Send Verification Code disabled when empty
- Dashboard loads as admin (sections visible, Quick Actions present)
- Dashboard loads as viewer (Quick Actions absent, mutating buttons disabled)
- Sidebar links visible for both roles (correct items, correct absence)
- Accounts page empty state
- Expenses page empty state (with no accounts: "Add Expense" disabled with tooltip)
- Income page empty state
- Users table loads (columns present)
- Projections page loads (controls visible)
- Pending Approvals page accessible by URL (platform admin configured — 403 potentially resolved)
- Viewer: mutating buttons disabled (aria-disabled pattern)
- Viewer: Export/Import absent from sidebar
- Unauthenticated access → redirect to `/login`
- Unknown route → redirect to `/dashboard`
- Mobile: sidebar collapsed by default, toggle button present
- Column headers visible (Accounts, Expenses, Income, Users tables)
- Settings dialog accessible via user menu

18.2 Fixture-Managed Tests (create own records, clean up in afterEach)

All grouped in `test.describe.serial()` blocks with `afterEach` cleanup:

- Account CRUD (create valid, create invalid/validation, edit, delete)
- Expense CRUD (create valid — no accrual, create valid — with accrual, create invalid, edit, delete)
- Expense row actions: Mark as Paid, Toggle Exclusion (include / exclude cycle), Duplicate
- Income CRUD (create valid, create invalid, edit, delete)
- Income row actions: Mark as Received, Duplicate
- Invite User (valid, duplicate username error, invalid email error)
- Change User Role
- Disable/Enable User
- Settings: update Site Details, User Details, Change Password, Budget Reminders toggle
- Account filter (requires two accounts + expenses/income spanning both)
- Search filter on Expenses / Income

18.3 Isolated-Sequence Tests (separate npm script, workers: 1, Mode 3 seed)

- Dashboard overview values match known seeded data amounts
- Quick Actions: Income Renewal → toast + data update
- Quick Actions: Expense Accrual → toast + data update
- Quick Actions: Budget Reminder → toast
- Quick Actions: Income Confirmation → toast
- Projections chart with seeded data — view selector options, period buttons, date filter
- Export: download `.export` file (admin)
- Import: upload `.export` file → data appears in UI (admin)

19. Deferred / Not Yet Explored

Area	Notes
Sign Up step 2 (verification code entry)	Requires real email or OTP test strategy (see PRD D7)
Password Reset step 2 (code + new password)	Same dependency as above
Approvals page with platform:manage permission	<code>e2e_admin</code> does not hold <code>platform:manage</code> ; need platform admin user or permission grant
Column sorting behaviour	Sort buttons confirmed present in table headers; sort direction change and API re-fetch not verified
Pagination behaviour	No pagination observed with current data volume; need 20+ rows to trigger
Projections view selector — exact option labels	Source not read in detail; recommend <code>src/features/projections/ProjectionsPage.tsx</code>
Income row action label verification	"Mark as Received" vs "Mark as Paid" label on income rows — confirm exact text from source
Settings dialog — which sections are viewer-accessible	Not tested with viewer; check <code>WithPermission</code> guards on each accordion panel
Error boundary fallback	<code>[role="alert"]</code> full-screen error — not triggered during exploration
<code>/logout</code> route behaviour	Route exists in code (<code>LogoutRoute</code> component); behaviour: calls <code>logoutManager.logout()</code> → redirect to <code>/login</code>

20. Recommended Product-Code Testability Improvements

These are minimal changes that would improve E2E locator reliability. They have no runtime behaviour impact.

Element	Gap	Recommendation
Quick Action cards	No accessible role or testid	Add <code>role="button"</code> or <code>data-testid="quick-action-{slug}"</code> to each card
Login inline error	No <code>[role="alert"]</code> or <code>testid</code>	Resolved — <code>role="alert"</code> added to the error <code><div></code> in <code>LoginForm.tsx</code>
Projections view selector	Labels unknown without code read	Verify label text in source; add <code>data-testid</code> if labels are dynamic

Appendix A: Data Strategy by Phase

Phase	Seed Mode	Data Present	Tests Applicable
Blank	Mode 1 (migrations only)	No site, no users	Not used by current tests
Baseline (Gate A)	Mode 2 (migrations + <code>baseline.sql</code>)	Site + 2 users, no financial data	All parallel-safe + fixture-managed
Seeded	Mode 3 (Mode 2 + import artifact)	Site + users + financial data	Isolated-sequence + projections + dashboard values

Appendix B: Form Sheet vs Dialog Pattern

Feature	Component	URL Change	Locator
Create/Edit Account	Centered modal dialog	Yes (route child)	<code>getByRole('dialog')</code>
Create/Edit Expense	Right-side sheet	Yes (route child)	<code>[data-slot="sheet-content"]</code>
Create/Edit Income	Right-side sheet	Yes (route child)	<code>[data-slot="sheet-content"]</code>
Invite User	Right-side sheet	Yes (route child)	<code>[data-slot="sheet-content"]</code>
Export dialog	Centered modal dialog	No (state-driven)	<code>getByRole('dialog')</code>
Import dialog	Centered modal dialog	No (state-driven)	<code>getByRole('dialog')</code>
Forgot Password	Centered modal dialog	No (state-driven)	<code>getByRole('dialog')</code>
Sign Up	Centered modal dialog	No (state-driven)	<code>getByRole('dialog')</code>
Settings	Centered modal dialog	No (state-driven)	<code>getByRole('dialog')</code>
Change Role	Centered modal dialog	No (state-driven)	<code>getByRole('dialog')</code>

Sheet vs dialog navigation note: When a right-side sheet is open and occupies the overlay, navigating away (`page.goto(url)`) is more reliable than pressing Escape to close it before the next action.

Appendix C: Radix Custom Control Interaction Patterns

```
// Custom combobox (Radix Select / Combobox) – NOT a native <select>
// DO NOT use .selectOption()
await page.getByLabel("Account").click(); // opens dropdown
await page.getByRole("option", { name: "Main Checking" }).click();

// Custom date picker
await page.getByRole("button", { name: "/Jan.*2026/" }).click(); // opens calendar
await page.getByRole("button", { name: "Today" }).click(); // select today
await page.getByRole("button", { name: "Accept" }).click(); // confirm

// Radix dropdown menu
await page.getByRole("button", { name: "Open menu" }).click(); // opens context
```

menu

```
await page.getByRole("menuItem", { name: "Edit" }).click(); // select action
```